

CS 58010: Scalable Learning Systems

Subject: CS Faculty: Faculty of Engineering and Natural Sciences

SU Credit: 3 , ECTS Credit: 10.00 / 10.00 ECTS

Instructor(s): [Kubilay Atasu](#)

Language of Instruction: English

Level of Course: Graduate

Planned Learning Activities: reading, presentation, discussion, and project prototyping

CONTENTS (ENGLISH)

This course provides a broad overview of state-of-the-art parallel and distributed machine learning (ML) and deep learning (DL) algorithms and systems, with a strong focus on the scalability, resource efficiency, data requirements, and robustness of the solutions. This course covers effective ways to map state-of-the-art ML and DL solutions to parallel AI accelerators such as GPUs and TPUs. A set of techniques are presented to efficiently scale ML and DL workloads to a large number of distributed machines in the presence of system failures and malicious attacks. Finally, methods for improving the scalability and efficiency of generative learning and graph learning approaches are covered.

Course topics include

- Overview of parallel and distributed ML/DL algorithms
- Performance and scalability of state-of-the-art ML/DL systems
- Hardware-accelerated ML/DL solutions
- Federated machine learning systems
- Scaling generative AI systems
- Scaling graph learning systems

İÇERİK (TÜRKÇE)

Bu ders, çözümlerin ölçeklenebilirliği, kaynak verimliliği, veri gereksinimleri ve sağlamlığına güçlü bir şekilde odaklanarak, son teknoloji paralel ve dağıtılmış makine öğrenimi (MÖ) ve derin öğrenme (DÖ) algoritmaları ve sistemlerine ilişkin geniş bir genel bakış sunar. Bu derste GPU'lar ve TPU'lar gibi paralel yapay zeka hızlandırıcılarına son teknoloji MÖ ve DÖ çözümlerini eşlemenin etkili yolları ele alınır. Sistem arızaları ve kötü amaçlı saldırılar varlığında MÖ ve DÖ iş yüklerini çok sayıda dağıtılmış makinaya verimli bir şekilde ölçeklendirmek için bir dizi teknik sunulur. Son olarak, üretken öğrenme ve grafik öğrenme yaklaşımlarının ölçeklenebilirliğini ve verimliliğini iyileştirme yöntemleri ele alınır.

REFERENCE BOOKS

[Dive into Deep Learning](#) by Aston Zhang, Zack. J. Lipton, Mu Li, and Alex. J. Smola.

OBJECTIVE

This course aims to provide the students with skills to build efficient, scalable, and robust parallel and distributed machine learning (ML) and deep learning (DL) solutions.

LEARNING OUTCOMES

1. Demonstrate deep understanding of parallel and distributed machine learning and deep learning algorithms and systems by analyzing and discussing research papers.
2. Design and implement scalable and efficient machine and deep learning algorithms and systems, evaluate time-space and cost-performance tradeoffs.
3. Analyze and evaluate the resiliency of federated machine learning algorithms and systems against various attacks.
4. Formulate research questions and objectives and relate them to the relevant literature in the broad field of deep generative learning systems.

ASSESSMENT METHODS AND GRADING

- 1) Group paper presentations (20%): each student group needs to choose two research papers from a given set, present them (15 minutes), and lead the discussion (5 minutes).
- 2) Group research project (60%): The goal is to improve the speed, scalability, resilience, or accuracy of an existing ML/DL approach. The students need to hand in a project report in the style of a short scientific paper stating their contribution to the overall system performance. Grading: 40% for the project report and 20% for the project presentation and Q&A.
- 3) Homeworks (20%): written, oral, or both.

AI USE

Students are allowed to use AI tools such as ChatGPT, Gemini, or Claude to support their understanding of course concepts and to assist in editing and summarizing their own work. However, AI tools may not be used to generate full-length written content or arguments for graded assignments. AI may be used for language refinement but not idea generation.

For the final paper, AI use is limited to editing only after a full draft is written by the student. Any use outside these boundaries will be considered a violation of the course policy. Students do not need to disclose AI use that aligns with these rules. However, if use goes beyond the permitted scope, they must consult the instructor in advance and provide documentation.

Students may not upload course materials, including slides, lecture notes, or assignment prompts, to any AI platform. If you are unsure whether a certain use is acceptable, contact the course instructor by email or during office hours.